Note of Fast Accurate Eigenvalue Computations Using The Cholesky Factorization

Theorem 2.3:
Let $G = BD \in \mathbb{R}^{n\times n}$ be nonsingular and let $\hat{G} = (B + \Delta B)D$ where $\|\Delta B\| \le \sigma_n(B)$.

$$\left|\frac{\sigma_i(\hat{G}) - \sigma_i(G)}{\sigma_i(G)}\right| \le \frac{\|\Delta B\|}{\sigma_n(B)}, \quad i = 1, \ldots, n.$$

Co

$$G = L\tilde{D}^{-1}\tilde{D}$$
$$\tilde{G} = (L\tilde{D}^{-1} - \Delta\tilde{D}^{-1})\tilde{D} = L_{(1)}\tilde{D}^{-1}\tilde{D}$$
$$\left\|L\tilde{D}^{-1}\right\| \le \frac{\left\|\Delta\tilde{D}^{-1}\right\|}{\sigma_n(L\tilde{D}^{-1})}$$

1 Derivative of eigenvalues and eigenvectors

Derivative of an eigenvalue $\lambda(X)$ and an eigenvector $u(X)$ of matrix X at X_0.

$\cdot^+$ means Moore-Penrose inverse.

TODO:

1. Relation of these two expressions?
2. SVD when has non-simple roots?
3. Derivative of normalization operation?

1.1 Real symmatric case

$X_0 \in \mathbb{R}^{n\times n}$, $X_0 = X_0^T$:

$$X_0 u_0 = \lambda_0 u_0$$

and λ_0 be a simple eigenvalue, then

$$Xu = \lambda u, \quad u^T u = 1, \quad X \in N(X_0)$$

$$\mathrm{d}\lambda = u_0^T(\mathrm{d}X)u_0$$
$$\mathrm{d}u = (\lambda_0 I_n - X_0)^+(\mathrm{d}X)u_0$$

1.2 General case

1.2.1 Normalization $u_0^* u = 1$

See [2]

$Z_0 \in \mathbb{C}^{n\times n}$:

$$Z_0 u_0 = \lambda_0 u_0$$

and λ_0 be simple eigenvalue, then

$$Zu = \lambda u, \quad u_0^* u = 1, \quad Z \in N(Z_0)$$

$$\mathrm{d}\lambda = \frac{v_0^*(\mathrm{d}Z)u_0}{v_0^* u_0}$$

$$\mathrm{d}u = (\lambda_0 I - Z_0)^+ \left(I - \frac{u_0 v_0^*}{v_0^* u_0}\right)(\mathrm{d}Z)u_0 \tag{1}$$

where $Z_0^* v_0 = \bar{\lambda}_0 v_0$, i.e. v_0^* is the left eigenvector of Z_0 corresponding to λ_0.

1.2.2 Normalization diagC=0

See [1]

Assume: $A \in \mathbb{C}^{n\times n}$?, $\lambda_i \neq \lambda_j \ \forall i \neq j$. $AU = U\Lambda$.

$$\mathrm{d}\Lambda = I \circ \left(U^{-1}\mathrm{d}A\, U\right)$$

Normalization: $\mathrm{diag}\left(U^{-1}\mathrm{d}U\right) = 0$:

$$\mathrm{d}U = U\left(F \circ \left(U^{-1}\mathrm{d}A\, U\right)\right) \tag{2}$$

where $F_{ij} = (\lambda_j - \lambda_i)^{-1}$ for $i \neq j$, and zero otherwise.

In general, Eq.(1) is different from Eq.(2). Eq.(2) seems has problem with complex conjugate roots, also it needs extra normalization step to get accurate result.

2 Derivative of SVD

See [1].

Assume $A \in \mathbb{R}^{m\times n}$,

$$A = USV^T,$$

then

$$\mathrm{d}S = I \circ \left(U^T \mathrm{d}A\, V\right).$$

Assume $m \leq n$.

Defined $\mathrm{d}C = U^{-1}\mathrm{d}U$, $\mathrm{d}D = V^{-1}\mathrm{d}V$ (note $\mathrm{d}C$ and $\mathrm{d}D$ are anti-symmetric), and partition

$$S = \begin{bmatrix} S_1 & 0 \end{bmatrix}, \quad \mathrm{d}S = \begin{bmatrix} \mathrm{d}S_1 & 0 \end{bmatrix}, \quad \mathrm{d}D = \begin{bmatrix} \mathrm{d}D_1 & -\mathrm{d}D_2 \\ \mathrm{d}D_2^T & \mathrm{d}D_3 \end{bmatrix}$$

$$U^T \mathrm{d}A\, V = \begin{bmatrix} \mathrm{d}P_1 & \mathrm{d}P_2 \end{bmatrix}$$

then

$$\mathrm{d}D_2 = S_1^{-1}\mathrm{d}P_2$$

$$\mathrm{d}C = F \circ \left(\mathrm{d}P_1 S_1 + S_1 \mathrm{d}P_1^T\right)$$

$$\mathrm{d}D_1 = F \circ \left(S_1 \mathrm{d}P_1 + \mathrm{d}P_1^T S_1\right)$$

where $F_{ij} = \left(s_j^2 - s_i^2\right)^{-1}$ for $i \neq j$ and zero otherwise.

$$\mathrm{d}U = U\left(F \circ \left(\mathrm{d}P_1\, S_1 + S_1 \mathrm{d}P_1^T\right)\right)$$

$$\begin{bmatrix}\mathrm{d}V_1 \\ \mathrm{d}V_2\end{bmatrix} = V \begin{bmatrix}\mathrm{d}D_1 \\ \mathrm{d}D_2^T\end{bmatrix}$$

References

[1] Mike Giles. An extended collection of matrix derivative results for forward and reverse mode algorithmic differentiation. page 23.

[2] Jan R. Magnus. On Differentiating Eigenvalues and Eigenvectors. *Econometric Theory*, 1(2):179–191, August 1985.